

Development of a Compact Vision-Based Tactile Sensor for Robotic End Effectors

Submitted by

LIU Zhaolin

Department of Mechanical Engineering
The University of Hong Kong

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Student Name:	LIU Zhaolin
Supervisor:	Wei Liu
Moderator or Co-supervisor:	Dr. WEI Yan
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DECLARATION

I hereby declare that this dissertation represents my own work, except where due acknowledge is made, and that it has not been previously included in a thesis, dissertation or report submitted to this university or to any other institution for a degree, diploma or other qualifications.

Signed

Liu Zhaolin

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ABSTRACT

Once a robotic end effector contacts an object, its structure can obscure the external view, limiting direct observation of contact location, local geometry and load variation. Commercial tactile and force/torque sensors also face constraints in fingertip applications, particularly in package size, wiring and cost. This dissertation develops a compact vision-based tactile sensor based on the open-source 9DTact principle.

The prototype integrates a compact camera, separate decoder board, eight-LED illumination PCB, printed enclosure, camera-locating structure and protected flex-cable route. A detachable mould supports staged fabrication of a G50/G5/G25 interface. The transparent G50 layer transmits light and supports the stack, the soft translucent G5 layer concentrates contact deformation, and the black G25 top layer forms an opaque contact surface that suppresses ambient light.

Wide-angle distortion, vignetting and non-uniform illumination made automatic grid detection unreliable. A manual 7×9 calibration interface was therefore developed to select 63 control points in order. The images were rectified and cropped using the radial basis function method of the reference 9DTact implementation. A spherical indenter with a nominal radius of approximately 4 mm established the pixel-to-depth mapping. After processing screw-contact images, the reconstructed surface retained the overall profile and repeated thread features, providing qualitative shape validation.

Two complete sensor prototypes were assembled, and one sensor was coupled mechanically to an FT300 six-axis force/torque sensor. The platform provides the hardware path for synchronous data collection. A transfer-learning study design uses the 9DTact pretrained model for initialisation and customised sensor data for target-domain adaptation.

Keywords: vision-based tactile sensor; robotic end effector; three-layer silicone; geometric calibration; depth calibration; shape reconstruction; six-axis force/torque

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Chapter 1 Introduction

1.1 Background and motivation

Robotic systems are moving from pre-programmed motion in structured environments towards embodied operation in which perception, the mechanical body and action are closely coupled [1], [2]. Physical AI further combines visual models, robotic data and manipulation hardware, yet contact still introduces uncertainty through occlusion, object-pose changes, compliant-surface deformation and incipient slip.

Tactile sensing provides local feedback where the end effector interacts with the object, complementing external vision for contact confirmation, position assessment and grasp adjustment [3], [4]. The project therefore develops a compact tactile module that can be fabricated and calibrated and that provides a mechanical mounting interface within the restricted space near a robotic fingertip.

1.2 Tactile sensing for robotic end effectors

A robotic end effector must operate through several contact regimes. In the process of approach, the external visual system can estimate the attitude and free space of the object. At the first touch, the controller needs a reliable contact event; in the closing stage, the contact patch will gradually expand and the local surface will be deformed; and in the process of maintenance or operation, tangential motion, pressure distribution changes or rapid changes in the image may foreshadow the occurrence of initial sliding. Therefore, effective tactile sensors should not only provide a single threshold, but should retain spatial information that can support contact positioning, geometric recovery, and force/torque estimation after proper calibration or learning. Traditional tactile technology is usually thin in structure and responsive, while commercial six-axis force/torque sensors can provide a clear combined force load. Its deployment scheme needs to weigh factors such as array wiring, spatial sampling, packaging volume, rigid load path requirements and cost. Visual-based tactile sensing provides a different balance scheme: the internal camera can observe the flexible optical interface under controlled lighting conditions and convert the deformation into a high-density image; the same image can support multiple algorithms without redesigning the tactile unit array - but this advantage depends on the camera and lighting device. , the careful integration of shell, elastomer and calibration process. The sensor is essentially an optical machine integrated system, rather than simply placing the camera behind the silicone back panel.

These dependencies are especially restrictive at a fingertip, where package depth, decoder access, flex-cable clearance and mounting features compete for the same limited volume. Moving one item can alter the optical working distance or cable bend, so a component substitution must be assessed across the sensor and followed by calibration after assembly.

1.3 Research problem

The open-source 9DTact platform combines compact dark-field imaging, shape reconstruction and image-to-six-axis-force/torque mapping and therefore provides a direct reference. The new camera, separate decoder and flex cable nevertheless change the package, optical geometry and calibration conditions, while the multilayer tactile interface must be fabricated with the available laboratory tooling. The work consequently redesigns the camera-LED arrangement, enclosure, cable route, silicone structure and calibration process and incorporates the FT300 mechanical interface. The gripper-integration section uses a

SolidWorks assembly of a self-developed two-finger gripper to illustrate the mechanical arrangement of the tactile sensors at the fingertips.

The core problem is how to translate the 9DTact principle into a compact, manufacturable and calibratable module using a new imaging chain and customised mechanical structure. The design must coordinate the camera and illumination, enclosure and cable route, mould and silicone stack, geometric and depth calibration, and shape reconstruction. The prototype is assessed through engineering records and qualitative contact geometry, with a target-domain adaptation study design based on the 9DTact pretrained model.

1.4 Aim and objectives

This dissertation aims to design, fabricate, calibrate and preliminarily validate a compact vision-based tactile sensor for robotic end effectors, while establishing the mechanical platform required for six-axis force/torque learning.

Objective 1 is to adapt the 9DTact optical principle to an OV5640 camera, separate decoder board and eight-LED illumination PCB. Objective 2 is to design the printed enclosure, camera-locating structure and protected cable route. Objective 3 is to develop a detachable mould and laboratory process for the G50/G5/G25 interface. Objective 4 is to implement manual 7×9 control-point selection, RBF mapping and sphere-based depth calibration. Objective 5 is to check shape reconstruction using structured contact geometry. Objective 6 is to establish the FT300 load path, synchronised acquisition and frame alignment, and to define a small-sample transfer-learning study design. A SolidWorks model of the self-developed two-finger gripper separately illustrates fingertip integration.

1.5 Main engineering contributions

First, the compact camera, separate decoder board, eight-LED illumination PCB, printed enclosure, camera-locating feature and flex-cable protection were integrated into one design. Second, a G50/G5/G25 three-layer silicone interface and detachable casting mould were developed, together with a fabrication process covering mixing, vacuum degassing, staged casting, curing and demoulding.

Third, a manual 63-point geometric-calibration interface was developed for the new wide-angle camera and combined with RBF rectification, effective-region cropping and sphere-based depth calibration. Fourth, two sensor prototypes, screw-shape reconstruction and the sensor–FT300 mechanical platform were completed, together with an implementation plan for synchronised acquisition and target-domain transfer; model accuracy was not evaluated.

1.6 Dissertation structure

Chapter 1 introduces the research background, problem, objectives and engineering contributions. Chapter 2 reviews robotic tactile sensing and representative vision-based tactile sensors. Chapter 3 presents the design requirements, system architecture and development workflow. Chapter 4 describes the hardware design and fabrication, while Chapter 5 covers geometric calibration, depth calibration and shape reconstruction. Chapter 6 presents the FT300 platform, synchronised acquisition and frame alignment, and a transfer-learning framework for limited target-domain data. Chapter 7 summarises the prototype and validation outcomes, compares the customised implementation with the original 9DTact, and briefly presents the mechanical assembly of the tactile sensors on a self-developed gripper. Chapter 8 presents the main conclusions and project summary.

Chapter 2 Related Work

2.1 Robotic tactile sensing technologies

Robot tactile sensors convert contact into electrical, magnetic or optical signals. Resistance and piezoresistive sensors infer loads through resistance changes, which can be made into a thin film form, but their accuracy may be affected by lag effect, drift and crosstalk; capacitive sensors realize detection by measuring changes in the geometric structure of the electrode. Although they have high sensitivity, in compact systems, parasitic capacitance, The shielding effect and array reading problems have become particularly critical; piezoelectric sensors are especially suitable for dynamic event detection, because their loaded materials can generate charges, and quasi-static contact requires additional design considerations; magnetic tactile sensors embed magnets or magnetic materials into flexible substrates and infer deformation through magnetic field changes, so that It realizes the effective separation between the replaceable surface and the sensing electronic components [8]. These sensing mechanisms have been reviewed in detail in the robotics literature [5] [6] [7], and their advantages depend on the specific task requirements: thin tactile unit arrays may be the most suitable for distributed contact detection, while high-bandwidth components are more suitable for vibration detection. The commercial six-axis force/moment sensor is located at the other end of the sensing chain. It can report the synthetic moment transmitted by the calibrated structure, which is of great value as a reference, but it cannot directly reveal the contact position or the shape of the contact spot; the visual-based tactile sensor uses the image of the deformation interface for detection. This will increase the packaging depth and calibration workload, but it can provide intensive spatial observation data and support the implementation of new perception output in the software. Table 2.1 summarises the principal technologies and their engineering trade-offs.

Table 2.1 Comparison of major robotic tactile-sensing technologies. Data compiled from the cited literature.

Technology	Primary signal	Engineering strengths	Deployment constraints
Resistive / piezoresistive	Resistance change	Thin, simple readout and scalable	Hysteresis, drift, cross-talk and array wiring
Capacitive	Capacitance change	High sensitivity and low static power	Parasitics, shielding and electromagnetic interference
Piezoelectric	Charge under dynamic stress	High bandwidth and dynamic sensitivity	Limited quasi-static response and charge conditioning
Magnetic	Magnetic-field variation	Compliant surface and separable transduction	Magnetic interference and configuration-specific calibration
Commercial six-axis F/T	Net force and torque	Calibrated six-channel wrench and defined frame	Resultant load only; rigid load path and package volume
Vision-based tactile	Image of a deforming interface	Dense spatial information and software-extensible outputs	Optical cavity, illumination, calibration and image processing

2.2 Representative vision-based tactile sensors

Retrographic sensing demonstrated that a camera, controlled lighting and a reflective compliant surface could recover detailed texture and shape [9]. GelSight developed this idea into high-resolution robot tactile sensors that estimate local geometry and force-related cues [10]. Later systems also used the spatial image to detect slip and refine manipulation [11]. These systems established the principal advantage of optical touch: a mechanically simple contact surface can produce a rich measurement when the optical path is controlled.

Compact packaging became a major design objective as tactile sensing moved towards robot fingers. GelSlim placed the camera and reflective gel in a finger-like geometry [12],

while GelSlim 3.0 combined shape, force and slip measurement in a refined compact package [13]. DIGIT emphasised a compact, manufacturable and cost-oriented module for learning-based manipulation [14]. TacTip follows a different morphology: an internal camera tracks the motion of pins connected to a soft biomimetic skin, avoiding reliance on a continuous reflective coating [15]. FingerVision combines visual and tactile information within the finger, illustrating how the contact camera can also support manipulation-oriented perception [16]. ReSkin represents a magnetic alternative with a replaceable soft skin and distributed magnetic response [17].

Dark-field optical sensors are particularly relevant to this project. DTact directly recovers high-resolution three-dimensional geometry from deformation under a dark-field configuration [18]. 9DTact extends this route with a compact multilayer sensor, shape reconstruction and a six-output force/torque regressor, supported by public code and data [19]. Other compact devices combine shape and force reconstruction through dense colour patterns or alternative optical arrangements [20], [21].

Table 2.2 Representative vision-based tactile sensors and the customised implementation. Data compiled from the cited literature.

System	Optical/contact morphology	Representative capability	Design relevance
GelSight	Reflective elastomer with controlled lighting	High-resolution geometry and force cues	Establishes camera-based contact imaging
GelSlim / GelSlim 3.0	Compact reflective gel finger	Shape, force and slip	Demonstrates fingertip-oriented packaging
DIGIT	Compact optical tactile module	High-resolution contact images	Emphasizes manufacturability and compactness
TacTip	Compliant skin with internal pins	Marker displacement and contact control	Alternative optical morphology
DTact	Dark-field compliant interface	Direct high-resolution 3D geometry	Precursor to the selected reconstruction route
9DTact	Compact dark-field multilayer sensor	3D reconstruction and six-axis regression	Provides open hardware, a public dataset and pretrained models
This work	G50/G5/G25 interface, compact camera and eight-LED PCB	Structured screw-contact reconstruction and FT300-ready platform	Practical adaptation for robotic end effectors

The comparison in Table 2.2 shows why 9DTact was selected as the starting point. The scheme provides an open and compact path from contact images to geometric parameters and six-axis force/torque estimation. At the same time, the published design scheme is a configuration scheme rather than a general suite. The camera's geometric parameters, lighting conditions, layer structure and calibration process are still coupled with each other, so replacing the camera or adjusting the packaging scheme needs to be re-engineered.

2.3 Shape reconstruction and force estimation

Shape reconstruction correlates image changes with surface deformation. Classical photometric stereo estimates surface orientation from multiple lighting directions [22]. Vision-based tactile systems may instead use colour gradients, markers, finite-element models or calibrated intensity-to-depth mappings. DTact and 9DTact employ a practical dark-field approach in which a reference image is subtracted, the valid sensing region is rectified and a calibration mapping converts pixel response into local depth. This route is attractive for a small

laboratory-built sensor because it separates geometric calibration from the learning problem and provides an interpretable intermediate depth map.

Force estimation is sensitive to the complete sensor configuration because the image depends on contact geometry, material response, friction, illumination and camera position. GelSlim research combined tactile images with inverse finite-element modelling for dense force estimation [23], while Helmholtz–Hodge decomposition has also been used to estimate contact force and torque [24]. 9DTact uses a DenseNet regressor with three force and three torque outputs; its public dataset and pretrained weights provide a basis for source-domain initialisation and cross-sensor transfer. Figure 2.1 shows the 9DTact shape-reconstruction and six-axis force/torque routes.

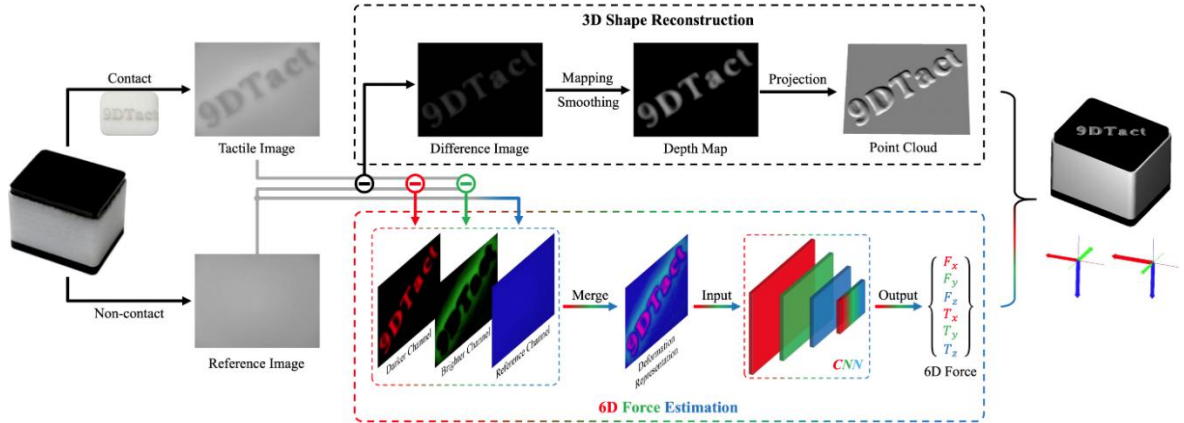


Figure 2.1 9DTact pipeline for three-dimensional shape reconstruction and six-axis force/torque estimation (reproduced from [19]).

Tactile images can also support grasp adjustment and incipient-slip detection [25]–[27]. These applications motivate end-effector integration but fall outside the validation scope of the present prototype. The completed work is limited to sensor fabrication, calibration, qualitative shape reconstruction and the FT300 mechanical load path.

2.4 Research gap and design opportunity

The existing literature provides powerful sensing principles and a number of compact reference design schemes, but the actual deployment still needs to rely on the engineering design of specific configurations. The camera selected in this project is different from the original 9DTact module in terms of size, decoder layout, field of view and distortion, which changes the working distance, center alignment and available image area. Flexible cables and external decoders also need to adopt a protective wiring scheme, which is not equipped in the reference package. Therefore, it is necessary to design a new shell, camera mount and LED layout.

Another engineering difficulty lies in the manufacture of three-layer tactile interfaces: it is difficult for homogeneous elastomers to achieve functions such as optical transmission, contact sensitivity, ambient light suppression and structural support at the same time. Although the G50/G5/G25 stacking structure can share the above functions, it needs to adopt a removable layered casting mold, and strictly control the leakage and bubbles; in addition, the calibration process also needs to be adjusted accordingly - strong deformation, green color deviation, vignetting and uneven contrast will weaken the automatic characteristics. Detection ability, therefore, the method of using orderly manual control points and radial basis function mapping has become a practical and easy-to-test method.

Geometric calibration can be regenerated from ordered control points for a new assembly, but learned force estimation transfers less directly. Its image-to-wrench relationship also

depends on camera response, illumination, elastomer mechanics and load-reference conventions. The publicly available 9DTact resources support source-domain initialisation, not a performance claim for the customised sensor.

Chapter 3 Sensor Design and System Methodology

3.1 Design requirements

The sensor is intended for installation at the front of a robotic finger or gripper and must therefore satisfy requirements for compact packaging, cable protection, fastener access, contact-load transmission and stable imaging. The design combines a compact camera and separate decoder board, an eight-LED illumination PCB, a printed enclosure, standard fasteners and cast silicone to form a low-cost prototype based on general-purpose components. Once assembled, the relative geometry of the camera, illumination and tactile interface should remain stable, while geometric and depth calibration must remain repeatable.

Table 3.1 summarises the design requirements, implemented responses and validation methods, distinguishing the completed assembly, imaging, calibration and qualitative-reconstruction checks from the unmeasured dimensions, lifetime, optical uniformity and batch-consistency metrics.

Table 3.1 Design requirements, implementation responses and validation routes.

Requirement	Design response	Validation method
Compactness	Compact camera, external decoder and integrated cable exit	Assembly and exploded-view inspection
Low-cost design using general-purpose components	Commercial modules, printed parts, standard fasteners and cast silicone	Component-level design review
Controlled imaging	Enclosed cavity, centred eight-LED PCB and black top layer	Qualitative image and reconstruction check
Mechanical stability	Dedicated camera fixation, housing datum and protected cable space	Assembly fit and handling check
Manufacturability	Detachable mould, seam sealing, degassing and staged casting	Completed process sequence
Calibration adaptability	Manual 7×9 control points, RBF mapping and sphere calibration	Rectified image and structured reconstruction
Modularity	Separable sensor head, decoder, FT300 fixture and self-developed gripper assembly model	Sensor-side interface, self-developed gripper assembly model and FT300 mounting inspection

3.2 Overall sensor architecture

As shown in Figure 3.1, the sensor comprises contact, imaging, mechanical and software layers. The G50/G5/G25 interface converts contact into observable deformation; the eight-LED illumination PCB, compact camera, flex cable and separate decoder board provide internal illumination and image acquisition; and the printed enclosure, camera-locating feature and cable outlet constrain the component positions.

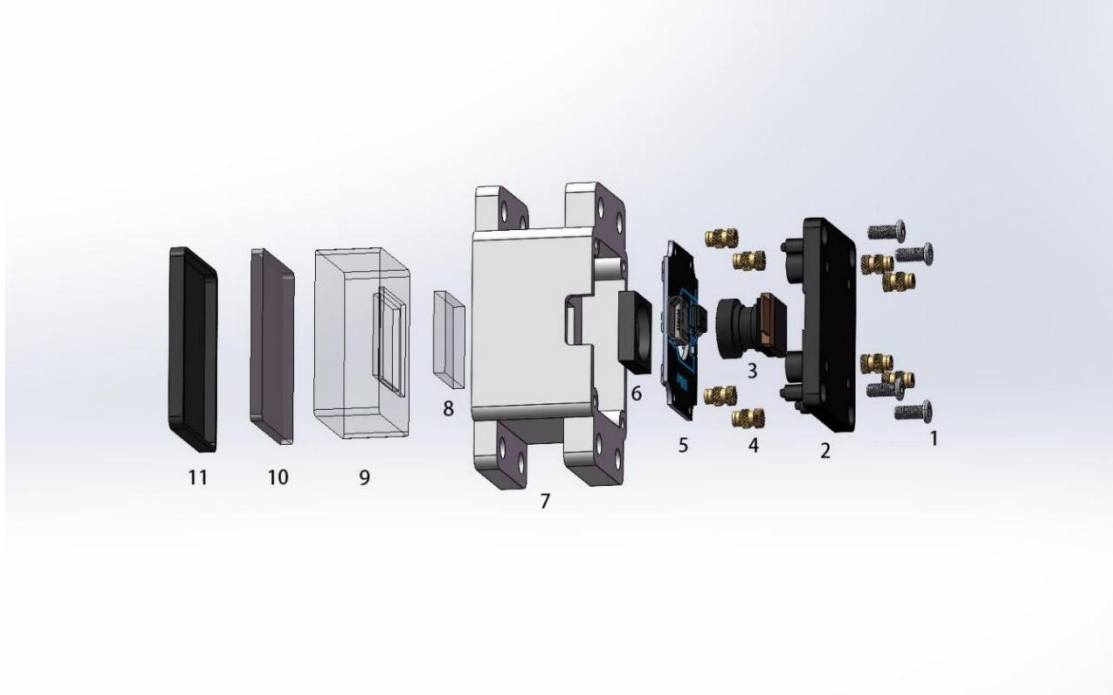


Figure 3.1 Exploded view of the customised vision-based tactile sensor.

1. Components: 1—M2×5 screw; 2—sensor base; 3—camera; 4—M2×5×3 heat-set insert; 5—LED board; 6—camera retaining ring; 7—sensor enclosure; 8—acrylic window; 9—transparent G50 silicone layer; 10—translucent G5 silicone layer; 11—black G25 silicone layer.

The software performs geometric rectification, effective-region cropping, reference-image subtraction and pixel-to-depth mapping. The force-estimation data structure defined in Chapter 6 uses rectified images as inputs and the six FT300 channels as labels, but no synchronised dataset was produced. The silicone pad, enclosure, external decoder and application-side fixture use separable interfaces to reduce disturbance of the camera-to-pad calibration relationship during peripheral maintenance.

The architecture separates the sensor core from the application-dependent mounting structure. The FT300 fixture lies outside the optical core, while four lateral lugs on the enclosure define the sensor-side mounting interface used in the self-developed gripper assembly. This arrangement reduces the risk of disturbing the calibrated camera-to-pad relationship when the external fixture changes.

The modular boundary also realizes the separation of verification tasks: the calibration file is still associated with the assembled sensor core, while the external fixture can independently check the suitability and cable gap.

3.3 Optical and electronic design

The sensor adopts a bottom-up internal imaging method: the camera is located under the silicone interface, and eight LED lights glow around the lens; the light passes through the G50 support structure into the G5 deformation layer, and the contact will change the light strength returned to the camera; the top layer of G25 can block ambient light and protect softer materials; the lens. The nominal field of view angle is about 160°; Figure 4.4 only shows the optical path and hierarchical structure.

The independent decoder board is arranged outside the sensor head to reduce the volume of the internal electronic components. The flexible cable is led through a special interface and protected through the gap, so as to avoid contact with the fastener and the main load path. This

layout can not only maintain the relative position between the camera, lighting PCB and the tactile window, but also ensure that the decoder is easy to connect and maintain.

The fixed geometric structure can support two processing paths: deep reconstruction will observe a consistent sensing area, and future force estimation can learn features from the same optical configuration.

3.4 Mechanical and material design

The printed shell can support the tactile interface, which is used to locate the camera and the lighting PCB, protect the internal components, and provide an external installation interface; its special positioning mechanism can constrain the lens position, and the independent cable wiring design can avoid the signal connection from bearing the positioning load. For details of the iterative optimization and manufacturing process of the structure, please refer to Chapter 4.

G50, G5 and G25 provide optical support, contact deformation protection and opaque protection functions respectively, matching material selection with imaging needs; for details of its layered structure and orderly preparation process, see Sections 4.6 and 4.7.

3.5 Development workflow

The development workflow is shown in Figure 3.2. It begins with component selection and measurement of the camera, decoder, LED PCB and cable constraints. CAD then defines the housing, fixing features, mould inserts and external interfaces. Printed prototypes are assembled with the electronics to check camera seating, LED alignment, cable clearance and access to fasteners. Structural observations are fed back into the next CAD iteration.

Mechanical fit was checked before moulding and calibration, so a displaced camera or obstructed cable was not mistaken for a software-mapping problem.

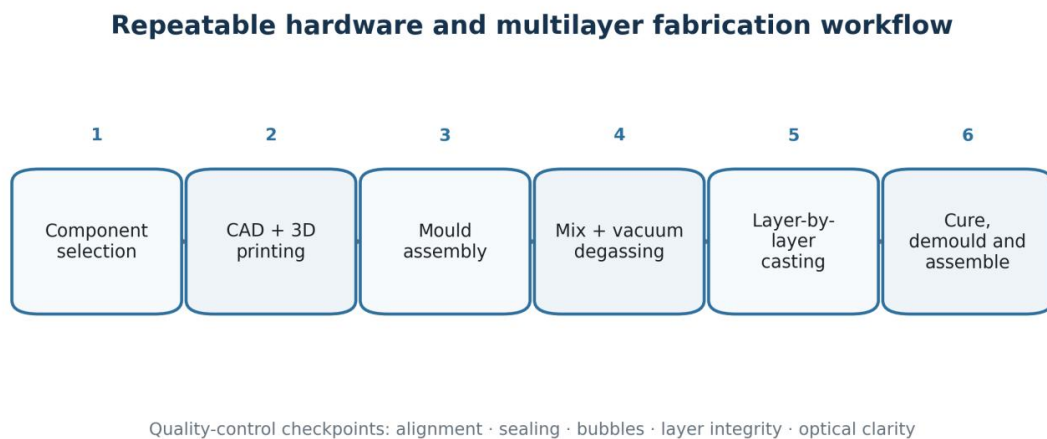


Figure 3.2 Development workflow linking component selection, CAD, fabrication, calibration and validation.

Once the mechanical layout was stable, a detachable mould was assembled from acrylic plates, threaded fasteners and printed inserts. After seam sealing, the silicone components were mixed and vacuum-degassed before staged casting and curing. The multilayer pad was demoulded, inspected and fitted to the enclosure. The optical assembly then underwent manual 7×9 geometric calibration, RBF-based rectification, effective-region selection and sphere-based depth calibration, followed by shape reconstruction as an end-to-end check. Finally, the

sensor was mechanically coupled to the FT300 and its load path was inspected, while the sensor-side mounting interface used for the self-developed gripper fingertip assembly was retained.

Chapter 4 Hardware Design and Fabrication

4.1 Adaptation from the original 9DTact design

The original 9DTact provides the sensing principle used here: an internal camera observes a deformable multilayer interface under controlled illumination, and calibration software converts contact images into geometry- or force-related outputs [19], [30]. The camera supplies dense spatial observations suited to a fingertip contact region, but camera pose, lens field of view, illumination, elastomer, enclosure and calibration are coupled. The present design retains the principle while adapting the LED-camera layout, locating features, internal clearance, sensing window and enclosure to a compact camera, separate decoder board and flexible cable. The cable outlet must also avoid fasteners and the main load path. The silicone interface is fabricated with detachable laboratory tooling, and calibration files are regenerated from the assembled prototype. This is an engineering-level adaptation to existing hardware, not a direct physical replication.

4.2 Camera, decoder board and LED PCB integration

The imaging link is assigned between the small camera head inside the tactile module and the independent decoder plate outside the main optical cavity. This layout reduces the space requirement that must be placed directly behind the flexible surface, and makes the larger interface electronic components still convenient to operate. This separation also brings additional mechanical requirements: the flexible cable must be led out of the sensor and must not be sharply folded, stuck by fasteners or exposed as structural components. Therefore, the decoder is regarded as a maintainable external module, while the position of the camera is constrained by the sensor shell.

There are eight LEDs on the lighting PCB, arranged around the central opening; the camera lens is placed in the opening, so that its optical axis is roughly located in the center of the tactile window, and the lighting light approaches the contact interface from multiple azimuth angles. Controllable internal lighting is a typical feature of camera-based tactile sensors (such as GelSight and Digit), because the observed signals depend on both deformation and lighting geometry [10] [14]. In the present design, the LED board also provides a compact mechanical reference around the lens. The camera, board and lower housing must therefore be aligned together rather than assembled as independent items.

Locating the lens within the LED aperture is more than packaging: a lateral shift changes the viewed region and illumination pattern, invalidating the stored calibration map. The housing therefore locates the camera and PCB directly, while the flex cable is given clearance rather than used for positioning.

Figure 4.1 shows the implemented relationship among the camera, eight-LED PCB, flexible cable and decoder. The photographs establish the physical integration and the centred lens arrangement. The relevant engineering result is that the camera head and illumination board fit within the sensor body while the decoder remains accessible outside the fingertip package.



Figure 4.1 Integrated arrangement of the camera, flexible cable, decoder board and eight-LED PCB.

4.3 Mechanical housing and cable-routing design

The printed enclosure performs four functions: it supports the silicone interface around the sensing window; establishes the camera and illumination-PCB positions relative to that window; protects the optical cavity from handling and external light; and provides the load-bearing and mounting features for connection to an experimental fixture or robotic finger. The housing is therefore part of the imaging system, because a change in camera position or pad support can alter the image even when the contacting object is unchanged.

A dedicated camera seating feature was introduced to prevent the lens from floating inside the enclosure during assembly. The seating geometry constrains lateral position and working distance, while the surrounding PCB and fasteners define the remaining stack. The flexible cable follows a separate internal clearance and exits through a slot in the lower part of the housing. This path can keep the cable away from the tactile window, thus reducing the possibility of pinching the wire when clamping the shell; at the same time, the design also allows the replacement or repositioning of the decoder plate without interfering with the casting pad.

Four sensor-side mounting lugs are incorporated into the outer housing and are visible in both the CAD views and completed prototypes. They provide the mechanical interface for the FT300 fixture and self-developed gripper fingertip mounting structure while separating the calibrated sensor head from the application-side mounting structure. The camera-to-pad geometry should remain unchanged when the external fixture is exchanged.

4.4 CAD development and structural iterations

CAD modelling was used to coordinate the optical stack, enclosure, camera seat, cable outlet, mould inserts and external mounting features before printing. The exploded views make the assembly sequence explicit: the tactile pad and its supporting elements occupy the object-

facing side; the housing locates the optical components; the LED PCB surrounds the camera; and the rear structure and fasteners close the module. This digital assembly was especially useful for identifying interference that would be difficult to correct after silicone casting.

Several printed enclosures and mould components were produced during development. Figure 4.2 records representative iterations in openings, wall geometry, access space and overall form, providing physical feedback on component fit, assembly access and the final locating arrangement.



Figure 4.2 Representative iterations of 3D-printed enclosures and tooling for component fit and assembly.

Additive manufacture was appropriate for this stage because changes to local clearances and mounting features could be implemented without remaking hard tooling. The final printed finished product still retains visible layered lines and the inherent dimensional tolerance of laboratory polymer printing; therefore, post-assembly calibration should be an integral part of the design method, not an optional modification step.

4.5 Multilayer mould design

The tactile interface cannot be manufactured through a single pouring process, because it is composed of a three-layer structure, and each layer plays a different role in optical and mechanical properties. Therefore, this study has developed a removable mold composed of acrylic plates, threaded rods or bolts, nuts and printed inserts. Among them, acrylic plates provide a wide and easy-to-visual positioning clamping surface; threaded fasteners can exert distributed compression force; and printed inserts are used to define local cavity and positioning features. The component can be disassembled and assembled between different processes and can be disassembled after curing, thus reducing the need for stretching or cutting the finished soft parts during the disassembly process.

The modular structure will form a seam at the junction of the plate and the insert. Uncured silicone may seep out through tiny gaps, especially when filling the mold in stages; before pouring, hot melt adhesive needs to be applied continuously along the contactable interface. The sealing step is a practical response to the leakage path, not a scheme aimed at maintaining structural adhesion inside the sensor; when removing the fixture, the sealing layer can be removed to retain the reusability of acrylic plates and threaded fasteners.

Leakage at an insert boundary can disturb the edge for the following layer and contaminate the interface. A removable hot-melt seal suited the detachable tooling because it conformed to printed-to-acrylic seams without permanently bonding the plates.

Figure 4.3 shows two assembled moulds, the relationship between transparent plates and printed components, application of hot-melt adhesive and the casting configuration after sealing. Using two mould assemblies allowed more than one tactile element to be processed within the same development cycle. The tooling is detachable laboratory manufacturing equipment that supports staged fabrication and visual inspection. Cavity dimensions, clamping torque and seal geometry were not quantified to precision-production requirements, so no conclusion is drawn regarding batch manufacturing accuracy.



Figure 4.3 Assembly, seam sealing and casting process for the detachable multilayer mould.

4.6 Three-layer silicone tactile interface

The silicone stack separates functions that are difficult to achieve with a single homogeneous elastomer. From the camera side to the contact side, the layers are the transparent G50 support, soft translucent G5 deformation layer and black G25 top layer. These names follow the project material records and are used consistently as layer identifiers; they are not independent hardness measurements made in this dissertation.

The sensor uses bottom-up internal imaging. The camera lies below the silicone interface and the physical eight-LED PCB illuminates the stack around the lens. The light passes through

the transparent G50 support into the G5 deformation layer, in which local contact will change the propagation path of the light and the brightness distribution returned to the camera. The top layer of the black G25 is located on the contact side, forming an opaque surface, which can block ambient light and protect the softer materials below. A lens with a nominal field of view angle of about 160° covers the main sensing area.

Figure 4.4 summarizes the three-layer structure, LED lighting path and camera field of view. The two LEDs and layer thicknesses shown are schematic and do not represent the physical eight-LED arrangement or measured dimensions. The interfaces must remain continuous because bubbles, contamination or poor bonding can modify light propagation, create non-contact artefacts in the image and alter load transfer.

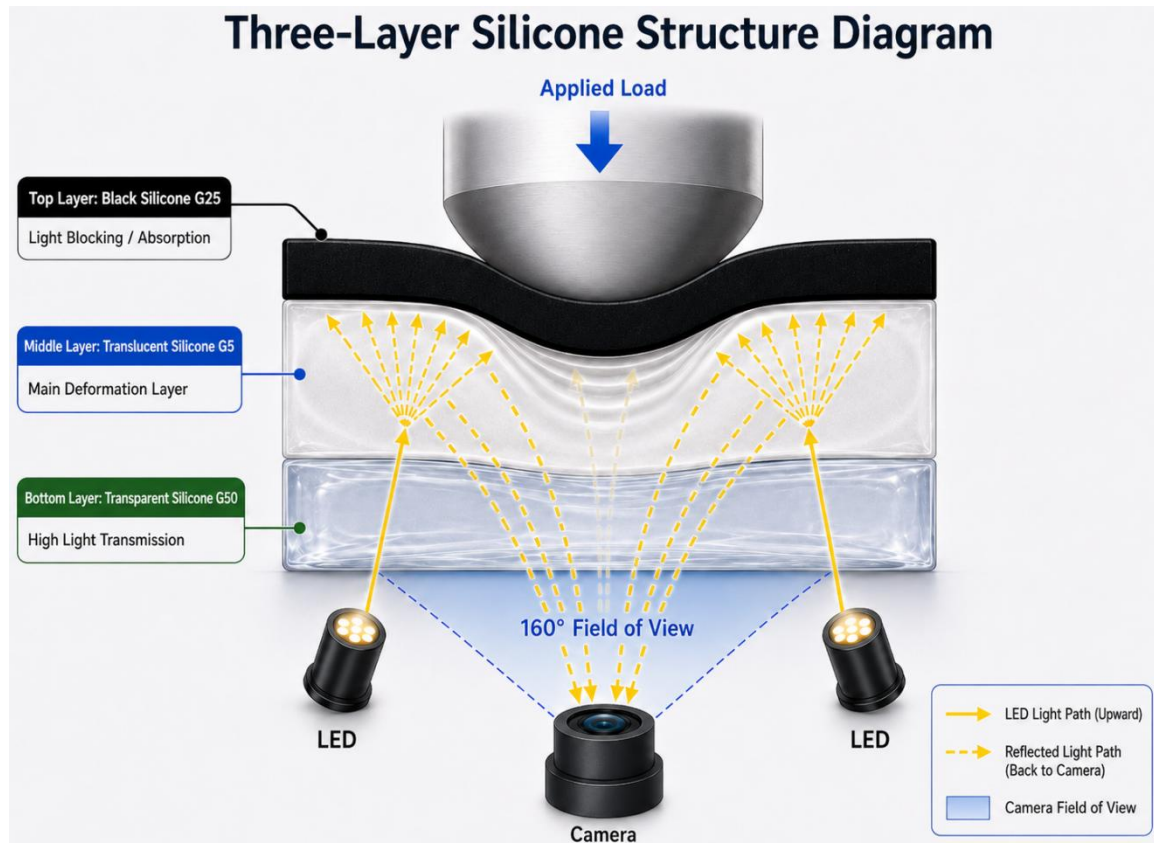


Figure 4.4 Schematic of the three-layer silicone interface, LED illumination paths and camera field of view (schematic; not to scale).

4.7 Silicone preparation and fabrication process

Fabrication began by preparing the mould components and checking that the printed inserts, acrylic plates and threaded fasteners could be assembled without obstructing the cavity. Contact surfaces were cleaned, the inserts were positioned, and the mould was clamped. Hot-melt adhesive was then applied along the plate and insert joints to close visible leakage paths. Performing sealing before mixing reduced the time-sensitive work required after the silicone components had been combined.

Each silicone was prepared by manually mixing its two components. The G50 packaging specifies an A:B ratio of 1:1 as the starting instruction for that layer. The mixture was then placed in a vacuum chamber connected to a pump so that entrained air could expand and leave the liquid before casting. This treatment reduces scattering voids in the transparent support and bubbles that could create false local features or modify compliance in the deformation layer.

The record confirms the two-step mixing and vacuum degassing process, but does not contain batch-level statistics on component mass, vacuum degree or processing time.

The pouring process is carried out sequentially to ensure that each material occupies its predetermined position. The transparent G50 layer constitutes the support structure and optical path facing the camera. When the layer has formed a form sufficient to accommodate the next pouring, the translucent G5 material is introduced as a flexible intermediate area. Finally, pour the black G25 layer to form an exposed contact surface. This layer-by-layer pouring method can make the previous layer complete the molding before the next layer solidifies, thus reducing the uncontrolled mixing phenomenon while maintaining the interface that can transmit deformation. This step-by-step pouring method is also convenient to check the coverage and can remove obvious surface pollutants between processes.

Degassing treatment has both optical and mechanical effects: the voids in G50 can still be observed in the process of subtracting the reference value, while the gaps in G5 may interfere with the local deformation transmission. Sequential casting kept G5 supported by the mould and G50 while the opaque surface was formed.

After curing, the nuts and plates were removed and the printed insert was separated from the multilayer pad. Detachable tooling reduces the tensile distortion imposed during demoulding, which is beneficial for the soft G5 region and for the bond between layers. The released pad was inspected for a continuous contact face, visible voids, edge damage and obvious misalignment before sensor assembly. Acceptance at this prototype stage was qualitative, based on whether the component could be integrated and imaged.

Figure 4.5 records material mixing, vacuum degassing, staged formation of G50, G5 and G25, and prototype assembly. The record shows that the three material functions were realised through a coherent laboratory fabrication process rather than by simply bonding unrelated prefabricated layers.



Figure 4.5 Three-layer silicone fabrication: (a) mixing; (b) vacuum degassing; (c) G50 casting; (d) G5 layer; (e) G25 layer; (f) final assembly.

4.8 Final sensor assembly

Final assembly combined the cured tactile pad with the printed enclosure, LED board, camera and cable system. The pad was seated at the object-facing opening so that the black G25 surface remained exposed and the transparent G50 side faced the optical cavity. The camera was placed in its dedicated seat, aligned through the central aperture of the eight-LED PCB, and retained by the rear structure. The flexible cable was routed through the internal clearance and lower exit before the housing was closed, after which it was connected to the external decoder board.

Two complete prototypes are visible in Figure 4.6. Both show the black tactile face, white printed housing, lower cable exit and separate decoder. The associated exploded views clarify that the tactile, optical and fastening components are arranged along a compact axial stack. They also show that the external mounting lugs belong to the sensor housing rather than to the silicone mould. The completed modules confirm mechanical compatibility among the redesigned subsystems and provide the physical basis for the calibration and shape-reconstruction work in Chapter 5.



Figure 4.6 Fully assembled vision-based tactile sensor prototypes.

4.9 Engineering challenges and implemented solutions

Fabrication first addressed mould leakage and trapped air. The joints between the acrylic plates and printed inserts were sealed continuously with hot-melt adhesive, and the mixed silicone was vacuum-degassed before sequential casting and curing of G50, G5 and G25. Functional separation prevented one material from having to provide support, deformation and light blocking simultaneously.

During assembly, a dedicated locating surface constrained the camera, while the cable outlet and protected clearance reduced sharp bending and pinching. Fit variation in the printed parts was corrected through successive enclosure and tooling iterations. These measures form a documented laboratory fabrication process. The assembled sensor then underwent the sensor-specific calibration in Chapter 5 to accommodate its actual camera-to-interface geometry.

The printed parts were assessed as an assembly because fastener access, the cable outlet, camera seat and optical stack compete for space. Successive CAD and print cycles resolved these interactions before casting, rather than deferring packaging problems to calibration.

Chapter 5 Calibration and Shape Reconstruction

5.1 Calibration challenge introduced by the new camera

Replacing the original 9DTact camera changes more than image resolution. Lens distortion, field of view, working distance, optical-axis position, illumination fall-off and the observed boundary of the silicone window all affect the correspondence between an image pixel and a location on the tactile surface. In the customised assembly, the raw calibration view is strongly curved near the perimeter, contains a green-dominant illumination field and becomes darker towards the corners. A mapping generated for a reference 9DTact sensor would therefore assign incorrect planar coordinates even if the same reconstruction code were used.

Calibrating the bare camera would leave assembly-dependent effects unresolved. Camera-to-pad spacing, alignment set by the locating features, and the boundary seen through the silicone stack all enter the image-to-surface relation. Using the entire head as a calibration object can bind the map to the geometric structure used for reconstruction, not to the nominal lens characteristics.

The assembled sensor is first geometrically corrected, and then deeply calibrated. The process cannot generate portable pinhole camera internal parameters, nor can it obtain the focal length, main point, radial distortion coefficient and tangential distortion coefficient separately; on the contrary, the distorted image is mapped to the regular grid by using a known dot matrix, and directly compensates for the composite generated by the lens, short optical path, tactile interface and assembly tolerance. Influence. Since the calibrated object is a complete imaging component, if the camera position changes, this calibration process must be repeated.

The official 9DTact repository provides the methodological starting point and sample configuration parameters, including the required camera resolution (640×480), frame rate (30 fps) and multi-frame average processing during calibration [30]. These values define the reference implementation scheme of 9DTact. The customized calibration screenshot verifies the normal operation of the real-time image acquisition function and the independent manual workflow, but does not confirm that the actual resolution, frame rate or exposure settings are exactly the same as the example. Therefore, this calibration is re-carried on a fully assembled custom sensor, rather than directly using the original configuration.

5.2 Manual 7×9 geometric calibration

The geometric target is a convex plate. When it is pressed against the tactile surface, a regular grid can be generated. The recorded calibration interface shows that the layout is 7×9 , and all 63 selected points are displayed in the specified order. The topology can provide a distributed correspondence in the central sensing area: after correction, each physical row and column should be linear and evenly spaced. The physical board photo and the numbered interface view are shown in Figure 5.1.

In the original 9DTact code, the point acquisition process is completed automatically: first, the reference image and the sample image are converted into grayscale images and the difference operation is performed; then the candidate contact points are screened through adaptive threshold processing and morphological closed operations; then the contour is filtered in area; calculate these contours The center of mass; finally, the obtained center points are arranged in the expected grid order [30]. When the target object produces 63 components with clear spacing and stable contrast, the method is more efficient. However, customized cameras will cause more significant wide-angle deformation, green lighting areas, dim surrounding

areas, and uneven appearance of the target. Under such conditions, fixed contours and sorting assumptions may lead to point merger, omission or misorder.

The actual adaptation scheme adopts the manual selection interface. The retained interface image records the title of "Manual 7×9 Calibration V2", the completion progress of 63/63, the point number superimposed on the surface grid, the magnification and undo controls, and the independent original preview and the corrected preview. These elements show how the operator can correct the missing or ambiguous positioning points before confirming the mapping results. The advantage of this interface is that it can ensure the robustness and inspectability of its operation even under complex imaging conditions.

Although the speed of manual selection is slower than contour detection, it can ensure the auditability of the point sequence. Each measurement point must correspond to a specific position in the ideal 7×9 grid; if any measurement point is omitted or misaligned, it may cause distortion in adjacent cells. By numbering the superimposed layer with the undo and magnification controls, the correspondence of each point can be verified before the map is generated.

The complete calibration process is listed in Table 5.1, while the interface diagram between the motherboard and the manual is shown in Figure 5.1.

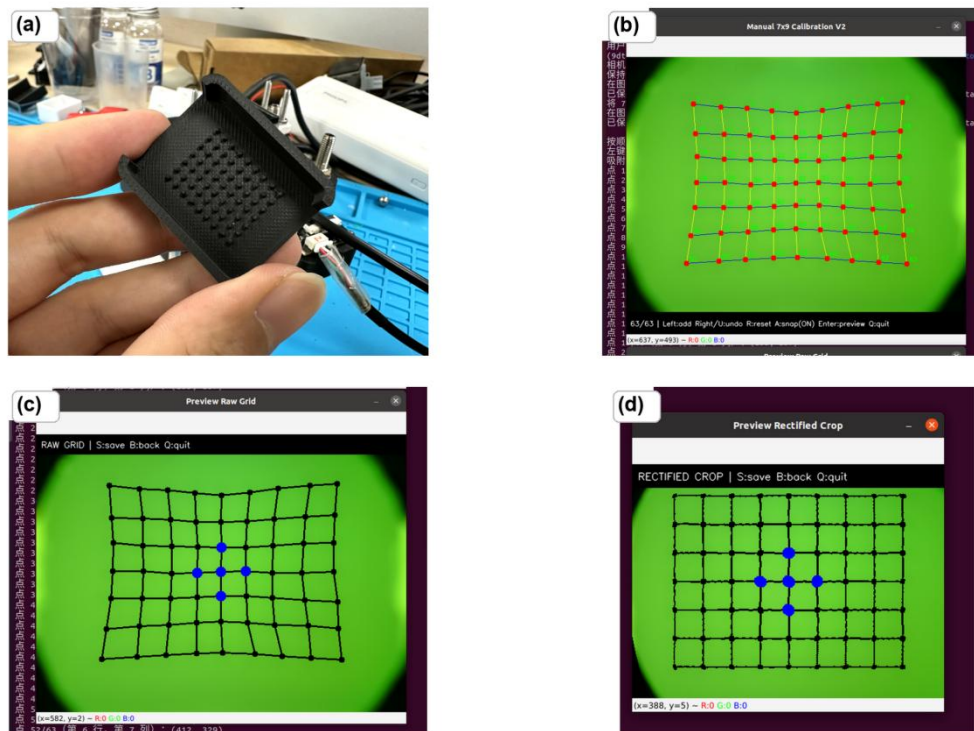


Figure 5.1 Manual geometric calibration: (a) raised-dot board; (b) 63-point selection; (c) grid before rectification; (d) rectified sensing region.

Table 5.1 Calibration stages, inputs, processing and outputs.

Stage	Input	Processing	Output
Control-point acquisition	7×9 raised-dot board	Manual ordered selection of 63 points	Measured point grid
Geometric rectification	Measured and ideal grids	Two-dimensional RBF interpolation	Row/column image maps

Stage	Input	Processing	Output
Region selection	Rectified image	Valid-bound and centre/crop selection	Consistent sensing window
Scale calibration	Known grid spacing	Pixel-distance conversion	Millimetres per pixel
Depth calibration	Reference frame and nominal 4 mm radius sphere contact	Difference image, contact boundary and spherical-cap geometry	Pixel-to-depth mapping
Shape reconstruction	Rectified contact image and depth mapping	Per-pixel conversion and surface rendering	Depth map and three-dimensional surface

5.3 RBF rectification and sensing-region selection

After point selection, the measured coordinates must be related to an ideal grid. The control points are defined in Equation (5.1):

$$p_i = (u_i, v_i), q_i = (x_i, y_i), i = 1, \dots, 63. \quad (5.1)$$

$$x = f_x(u, v), y = f_y(u, v). \quad (5.2)$$

The coordinate function in equation (5.2) can match these correspondences. The reference 9DTact implementation method constructs independent cubic radial basis function mappings for the rows and columns of the image, and evaluates these mappings on the whole image, thus generating a dense search table [30]. The custom interface records the RBF mapping stage and displays the corrected grid preview. Row and column mappings are sensor-specific calibration outputs; the project directory does not contain a unified archive of control points, physical grid spacing, crop definitions and complete mappings.

However, its visual effect is still very obvious: in the original preview, the nominally straight grid line will bend towards the image boundary, and the width of each cell is different; while in the corrected preview, the rows and columns in the central rectangular area have basically tended to be regular. Subsequently, the result of the geometric mapping will be combined with the selection of the sensing area. The cropping operation can remove the edge pixels with low support of the mapping coordinates in the control point domain, thus providing a consistent image size and center point for spherical-based depth calibration.

The crop is part of the calibration definition, not the presentation selection. The RBF interpolation function works best in the area surrounded by selected points; peripheral pixels will increasingly rely on extrapolation. Limiting the reconstruction process to the corrected center window can ensure that spherical fitting and depth finding are completed in the area where geometric correspondence has been established.

The retained interface report shows that the center point is near row 237 and column 319, the scale is 0.05949821 mm/pixel, and 32.84% of the pixels are outside the mapping boundary. These data come from the recorded calibration process; among them, the value of "32.84%" is a coverage diagnostic indicator, not a reprojection error, which does not mean that there is an error in the same proportion of reserved pixels; this value reveals why wide-angle images require clear cropping operations. The scale has not been independently verified by the standard manual calibration, and the sample scale or mapping file in the original 9DTact device cannot be used to replace the calibration process of customized sensors.

5.4 Sphere-based depth calibration

Geometric rectification supplies planar coordinates but does not determine indentation depth. Depth calibration uses an indenter with known geometry so that a contact image can be associated with a surface-height field. The project employed a metal sphere described in the project record as having a nominal radius of approximately 4 mm. Figure 5.2 shows the sphere together with the calibration interface, which contains reference (ref), loaded (img), difference (diff) and contact views. A fitted circular boundary is visible around the detected contact region.

For a sphere of radius R , physical contact radius a and radial distance ρ from the fitted centre, the spherical-cap height is calculated using Equation (5.3):

$$h(\rho) = \sqrt{R^2 - \rho^2} - \sqrt{R^2 - a^2}, 0 \leq \rho \leq a. \quad (5.3)$$

The sign may be reversed in software depending on whether indentation into the pad is represented as positive or negative. Converting the fitted radius from pixels to millimetres requires the geometric scale established in the preceding stage. The sphere thus links three quantities: contact position in the rectified image, physical surface geometry, and the grayscale change created by deformation.

The reference 9DTact configuration uses BallRad = 4.0 mm as an example [30]. The numerical agreement with the nominal project sphere supports use of the same spherical-cap geometry, while the customised mapping is computed from the new sensor's own rectified images, image scale and fitted contact region.

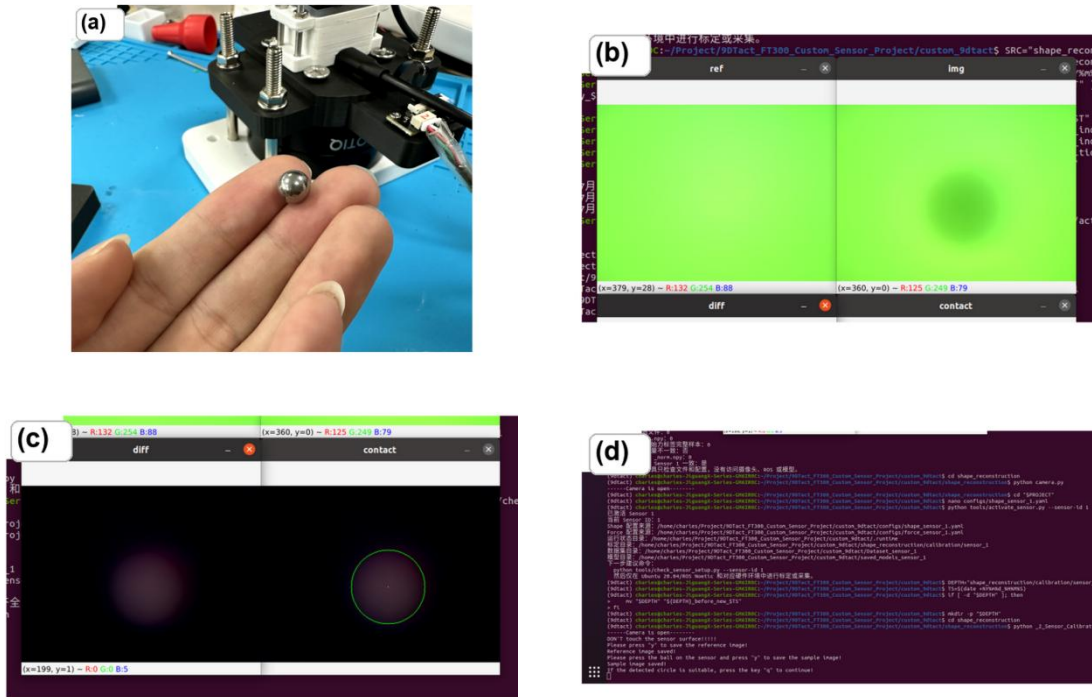


Figure 5.2 Spherical-indenter depth calibration: (a) metal sphere of nominal radius about 4 mm; (b) reference and loaded frames; (c) difference and contact region; (d) runtime state.

5.5 Pixel-to-depth mapping and reconstruction method

For a calibrated spherical contact, each pixel within the detected circle has a target height $h(\rho)$ determined by spherical-cap geometry. The intensity response at the same pixel is the difference between the rectified no-contact reference image $I_0(u,v)$ and contact image $I_t(u,v)$. Grouping pixels by greyscale difference and averaging their corresponding geometric heights yields a one-dimensional pixel-to-depth lookup function $L(g)$. This representation is applicable

to dark field sensors, because after geometric correction, the output value can be calculated independently at each pixel.

During the reconstruction process, the program first subtracts the current image from the reference image, suppresses the contactless background through threshold processing and truncation operations, and applies spatial smoothing to reduce the isolated intensity noise; then, the processed grayscale index is converted into a depth value through the calibration search function. The specific process is as follows (5.4) and (5.5) are shown.

$$g(u, v) = G(T[I_0(u, v) - I_t(u, v)]). \quad (5.4)$$

$$z(u, v) = L(g(u, v)) - L(g_0). \quad (5.5)$$

Here, $I_0(u, v)$ and $I_t(u, v)$ are the no-contact reference and current tactile images, respectively; T denotes thresholding and clipping, G denotes spatial smoothing, L is the lookup function obtained from depth calibration, and g_0 is the no-contact baseline index. The calibrated $\text{mm} \cdot \text{pixel}^{-1}$ scale converts pixel rows and columns to planar coordinates for dense surface generation. Figure 5.3 summarises the full sequence from rectified reference and contact images through differencing and lookup mapping to three-dimensional display. The method is calibration-based and does not use a neural network.

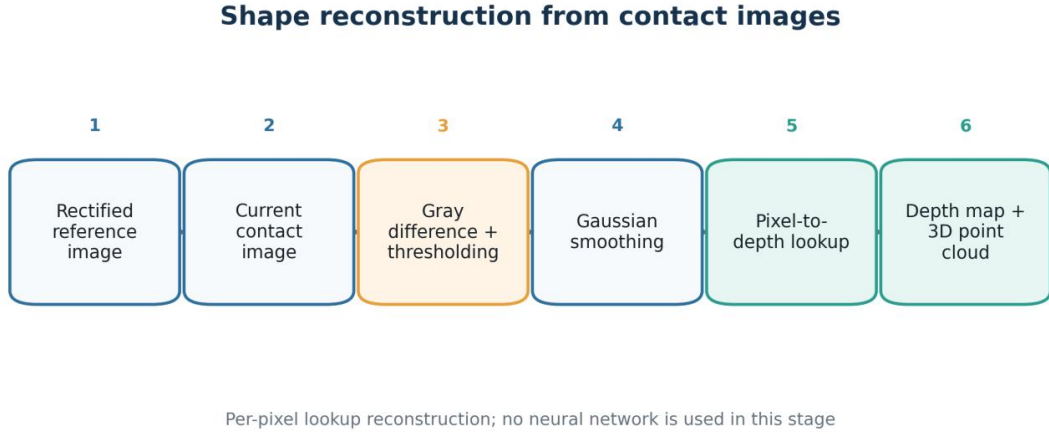


Figure 5.3 Pixel-to-depth shape-reconstruction workflow from rectified contact images to a depth map and three-dimensional surface.

The official reconstruction program uses a threshold parameter $\tau = 2$ in its example configuration, applies a 7×7 Gaussian operation before lookup, and stores the sensor-specific relation as `Pixel_to_Depth.npy` [30]. These settings describe the reference implementation; they do not prescribe identical values for every sensor. The customised system retains the same processing logic while using the manual geometric map and session-specific scale recorded in Section 5.3.

The lookup table is a calibration resource for specific sensors and sessions. Even if the contact point remains unchanged, changes in the reference coordinate system, exposure parameters, internal light or camera position will still change the grayscale difference input to $L(g)$; therefore, to convert the optical offset into apparent depth, the reference image and the calibration state must be recorded at the same time.

5.6 Shape-reconstruction result

The completed perception chain is evaluated by using screws to press against the tactile surface. Figure 5.4 comprehensively shows the amplified rendering results of the original

grayscale contact view, estimated depth map, three-dimensional surface and contact features. A faint slender response signal can be seen in the original image; in the processed depth map, the contact area and the background are clearly distinguished; and the surface view retains a series of transverse ridge sequences distributed along the axis of the object.

These repeated ridge lines are visually consistent with the thread structure and can achieve the function of transcending binary contact detection. This shows that the prepared optical layering, geometric correction and pixel-depth conversion functions can effectively retain the structural changes in a single contact plaque. The enlarged panel also clarifies why the flexible G5 layer and stable imaging geometry are crucial: after the beam passes through the support layer and smoothing, the local modulation effect of the contact surface still needs to be observed.

The result provides qualitative verification, indicating that the thread pattern and the overall screw outline can be effectively distinguished in the reconstructed surface. This paper does not quantitatively analyze spatial resolution, depth error and repeated positioning performance; see Section 5.7 for details of relevant discussion.

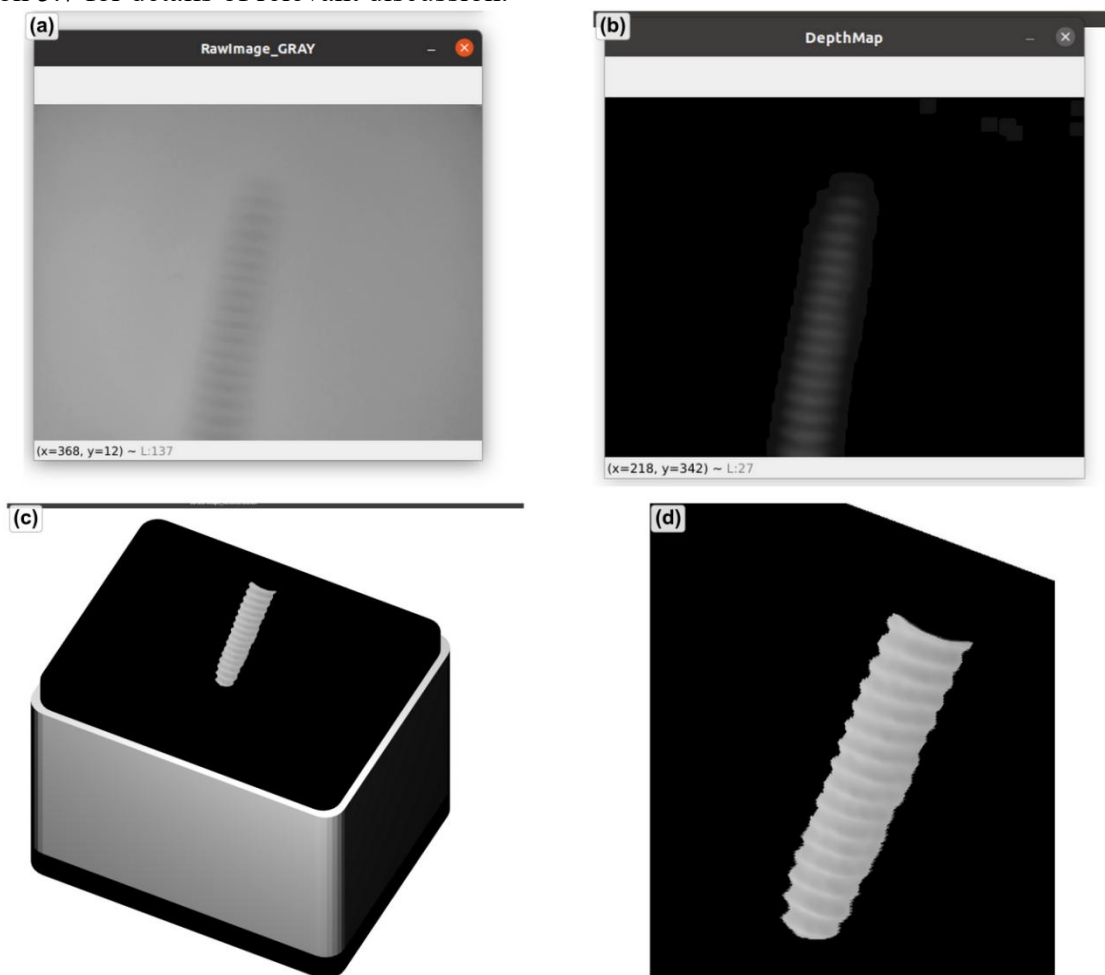


Figure 5.4 Screw-contact reconstruction: (a) contact image; (b) estimated depth; (c) reconstructed surface; (d) enlarged thread detail.

5.7 Discussion of calibration performance and limitations

Wide-angle distortion, vignetting and uneven light reference make automatic high-point detection unstable, so the manual 63-point interface provides control points that can be checked.

The RBF principle follows the original 9DTact implementation [30]; adjustments in practical applications involve point selection, cropping areas, image scaling and sensor-specific calibration. Spherical calibration and screw examples together verify the entire processing process from intensity response to structured contact surface.

The conclusion that thread reconstruction can support is more limited than the absolute accuracy test. After correction, smoothing and depth conversion, the repeated thread characteristics can still be observed, indicating that its spatial arrangement structure can be retained throughout the processing process. The result cannot determine the true depth of the groove, the minimum distinguishable features or the repeatability after reassembly; and these problems need to be solved with the help of traceable geometric data and repeated measurements.

The main uncertainties include manual point placement, grid and sphere size, camera movement during reassembly, RBF boundary extraposition, light and exposure changes, silicone thickness and viscoelasticity. Manual selection allows each control point to be detected under severe distortion, but the operator's error will be transmitted to the plane map; the spherical size and contact circle detection will affect the depth search. The records support functional checks and qualitative shape validation, while in-plane error, depth error and reassembly repeatability have not been quantified with independent spheres or step artefacts.

Chapter 6 Six-Axis Force Platform and Transfer-Learning Study Design

6.1 Customised sensor–FT300 experimental platform

The customised sensor was fixed above a Robotiq FT300 with a mounting fixture. The FT300 outputs F_x , F_y and F_z and torques T_x , T_y and T_z as the target-domain label reference [29]. Contact load passes through the silicone stack, printed enclosure and adapter, while the flex cable and decoder remain outside the main load path. The load path and standalone FT300 reading were completed, providing the hardware needed to collect synchronised images and six-axis labels.

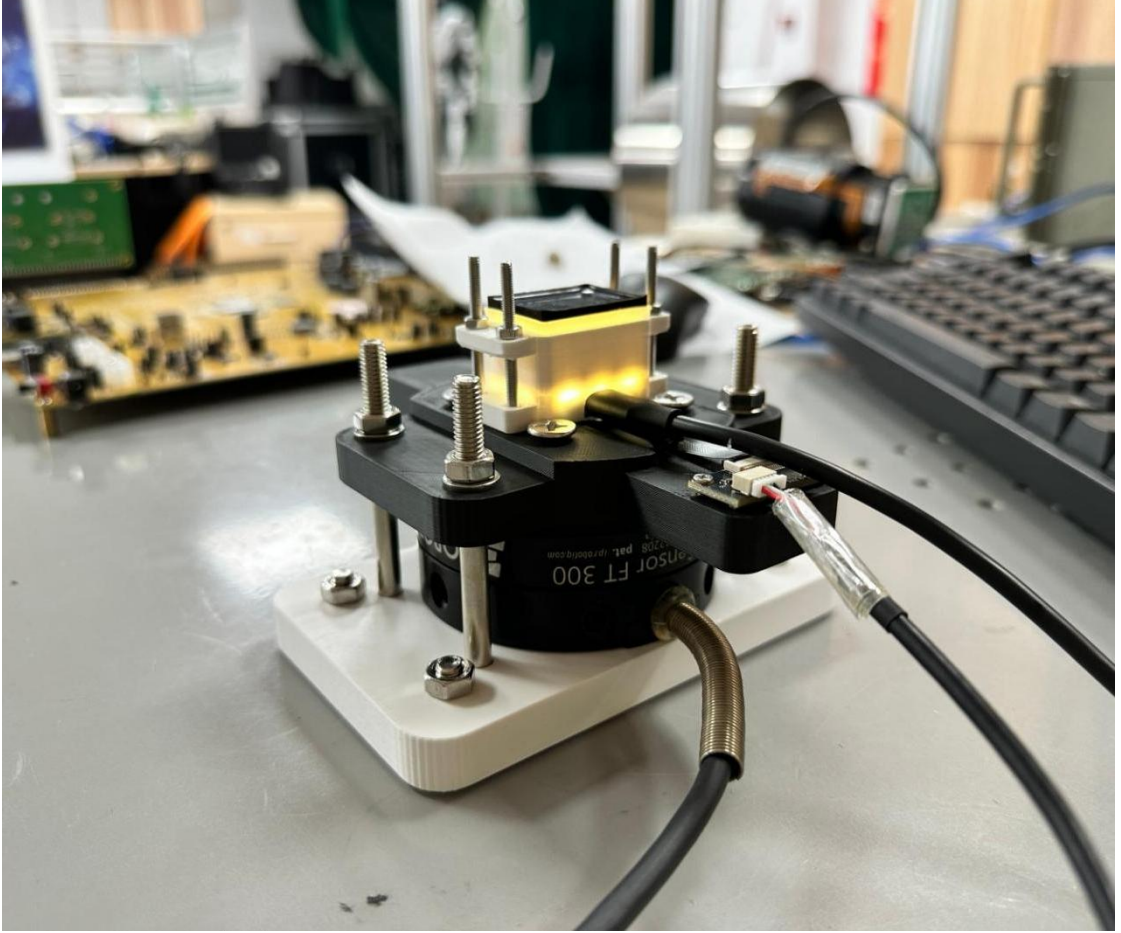


Figure 6.1 Mechanical coupling platform for the customised vision-based tactile sensor and FT300.

6.2 Synchronised data acquisition and wrench-frame alignment

Figure 6.2 shows the synchronised acquisition and quality-control structure. Each record should retain the rectified image, six-axis label, image and FT300 timestamps, trial identifier, zeroing state, coordinate transformation and calibration version so that mismatched or incompatible samples can be rejected.

Let F_s and τ_s denote force and torque in the source frame, and F_t and τ_t the corresponding quantities in the FT300 or target frame. If the frames differ only by rotation, force is transformed by Equation (6.1). If the torque reference points are also separated by r , Equation (6.2) includes the associated moment shift, where R rotates quantities from the source to the target frame.

$$\mathbf{F}_t = \mathbf{R}\mathbf{F}_s \quad (6.1)$$

$$\boldsymbol{\tau}_t = \mathbf{R}\boldsymbol{\tau}_s + \mathbf{r} \times (\mathbf{R}\mathbf{F}_s) \quad (6.2)$$

The 9DTact reference sensor and FT300 may use different axis directions, and different torque origins introduce the $r \times F$ term. Training data must therefore use a common component order, sign convention, N and N·m units, zero offset and torque reference point. These conventions should be resolved explicitly rather than inferred by a small-data model.

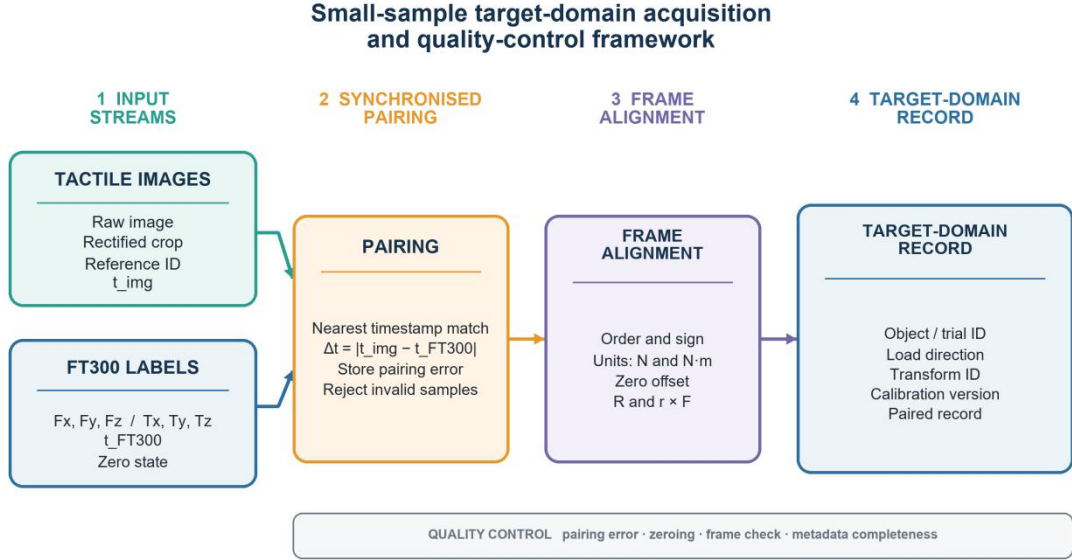


Figure 6.2 Framework for synchronised acquisition and quality control of customised tactile images and FT300 force/torque labels.

6.3 Transfer learning from the pretrained 9DTact model

The publicly available 9DTact dataset and DenseNet-169 weights provide source-domain initialisation [19], [28], [30], while synchronised customised sensor–FT300 records would form the target domain. Previous work reduced cross-sensor calibration data by freezing a tactile model and training a sensor-specific input layer [31]. Other optical-tactile studies adapted latent or image representations across sensors [32], [33]. These results motivate transfer, but they do not establish performance for the present prototype.

The sensor-to-sensor gap covers camera response, lens field of view and working distance, distortion, LED layout, crop region, silicone thickness and compliance, enclosure load path, force reference, coordinate axes and torque origin. These differences affect both image statistics and the deformation-to-load relation. Because the source dataset is much larger, direct concatenation could make optimisation follow the source distribution while leaving the customised sensor poorly calibrated. The study design therefore uses the public model to initialise visual features and target records to adapt the input representation, regression head and final backbone block.

Stage 1 uses the 9DTact DenseNet-169 weights to initialise early and intermediate visual features and replaces the original output with a new six-axis regression head. The weights provide a feature starting point; source-model outputs are not treated as quantitative results for the customised sensor.

Stage 2 trains the new regression head or a lightweight adapter with the backbone initially frozen. Target records then correct input differences, channel scale, zero offset and coupling. If zero-shot outputs retain useful correlation with FT300 labels, a linear scale-and-bias correction can be examined before unfreezing the backbone.

Stage 3 unfreezes the final dense block after validation performance stabilises and applies a smaller learning rate, early stopping and weight decay. This sequence updates fewer

parameters than retraining the entire network on a small dataset. Figure 6.3 summarises the three stages.

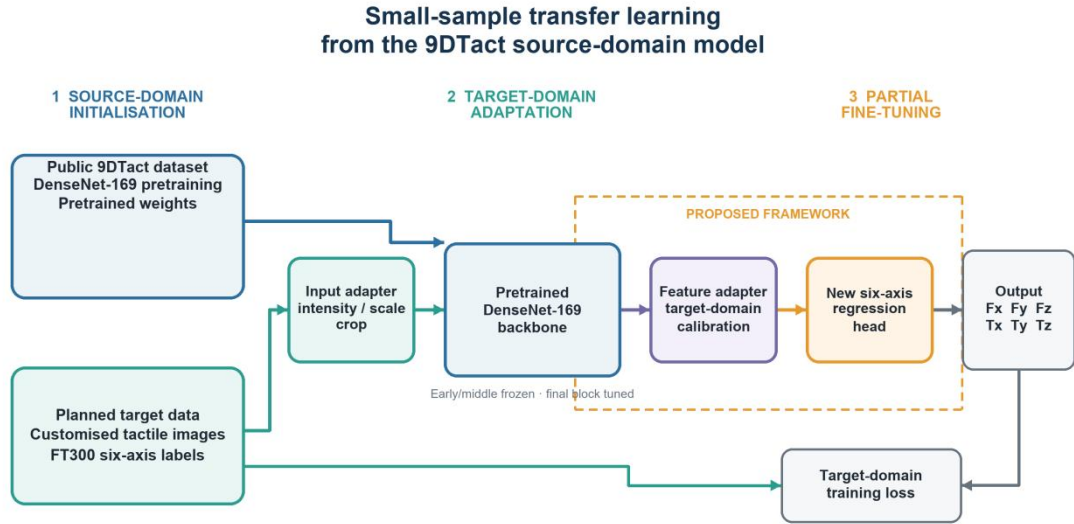


Figure 6.3 Transfer-learning framework using the public 9DTact source-domain model and a limited customised sensor–FT300 target-domain dataset.

6.4 Small-sample training and evaluation protocol

The protocol covers pipeline verification, target-domain adaptation and independent evaluation, with data and coordinates checked before model comparison.

A small set first checks image–label pairing, timestamps, coordinate directions, FT300 zeroing, data format and calibration version; it does not support an accuracy claim.

Training subsets then expand across central, edge and corner contacts; normal, positive and negative shear, eccentric and torsional loads; loading and unloading; and several geometries. Correlated adjacent frames are not independent samples.

Table 6.1 lists four planned model comparisons: zero-shot inference, training from scratch, regression-head retraining and adapter-based partial fine-tuning.

Table 6.1 Model-comparison protocol for small-sample six-axis force/torque estimation with the customised sensor.

Scheme	Initialisation	Trainable components	Purpose
A 9DTact zero-shot inference	Public pretrained weights	None	Zero-shot baseline
B Target-domain training from scratch	Random	Entire network	From-scratch baseline
C Pretrained backbone + new head	9DTact pretrained backbone	Six-axis regression head	Basic transfer
D Adapter + pretrained backbone + new head	9DTact pretrained backbone	Adapter, regression head and final dense block	Planned partial fine-tuning

Data splits must separate objects or complete trials and keep adjacent frames together. An independent test set should report channel-wise MAE, RMSE and R^2 in N and N·m. Compare Schemes B–D as target subsets grow; only consistent gains over Scheme B indicate lower labelling demand. No synchronised dataset is available, so this chapter presents a study design, not performance results.

Chapter 7 System Results, Gripper Integration and Discussion

7.1 Prototype and validation outcomes

Two vision-based tactile sensor prototypes were completed, integrating the compact camera, separate decoder board, eight-LED PCB, printed enclosure, camera-locating structure and protected cable route. The exploded view and assembly photographs record the component order and connections. Both prototypes have complete black tactile surfaces and externally accessible decoder electronics.

The detachable mould supports staged casting of G50, G5 and G25; the fabrication record covers sealing, mixing, vacuum degassing, curing, demoulding and assembly. Manual 63-point calibration, RBF rectification and sphere-based depth calibration were completed, and screw reconstruction retained the overall profile and repeated threads as qualitative shape validation. The sensor–FT300 platform established the load path; the transfer-learning design has not undergone performance validation.

7.2 Engineering comparison with the original 9DTact

The customised system retains the central 9DTact route in which an internal camera observes a multilayer tactile interface, the sensing region is rectified, and image changes are mapped to surface depth [19], [30]. The implementation is changed through the new camera and separate decoder board, eight-LED PCB, printed enclosure and locating surface, protected cable outlet, detachable-mould fabrication of the G50/G5/G25 stack, and manual 7×9 control-point calibration. Replacing the camera also changes the working distance, distortion distribution, effective image region and cable-space requirement, so the enclosure, relative illumination position and calibration files must be adapted together. Table 7.1 compares the resulting engineering differences between the two configurations.

Table 7.1 Engineering comparison between the original 9DTact configuration and the customised implementation.

Aspect	Original 9DTact route	Customised implementation
Camera / decoder	Original matched module	New compact camera with separate decoder
LED	Original illumination arrangement	Eight-LED PCB aligned with the new camera
Housing	Original package geometry	Redesigned printed enclosure and camera seating
Cable	Original routing	Dedicated outlet and protected internal clearance
Silicone	Original multilayer concept	G50/G5/G25 functionally separated stack
Mould	Original tooling route	Detachable acrylic/fastener/printed-insert mould
Calibration	Original automatic/custom routines	Manual 7×9 control points, RBF rectification and sphere depth calibration
Force reference	BOTA MiniONE in the public data route	FT300 mechanical experimental platform
Gripper interface	Application-dependent	SolidWorks assembly with two tactile sensors mounted on the fingertips of a self-developed two-finger gripper to demonstrate end-effector integration

Table 7.1 compares structure and implementation rather than performance. Original 9DTact reports quantitative reconstruction and learned six-axis force/torque results, whereas the present evidence comprises fabrication records, qualitative screw reconstruction and the

FT300 platform. The configurations and validation conditions differ and should be interpreted separately.

7.3 Simplified mechanical integration with a self-developed gripper

A simplified SolidWorks assembly of a self-developed two-finger gripper places two vision-based tactile sensors at the opposing fingertips, as shown in Figure 7.1. Their black surfaces face the grasping region, and connecting structures attach the housings to the finger ends. The model shows spatial arrangement, contact orientation and mounting only. No physical assembly, grasping, slip-detection or closed-loop-control test was conducted.

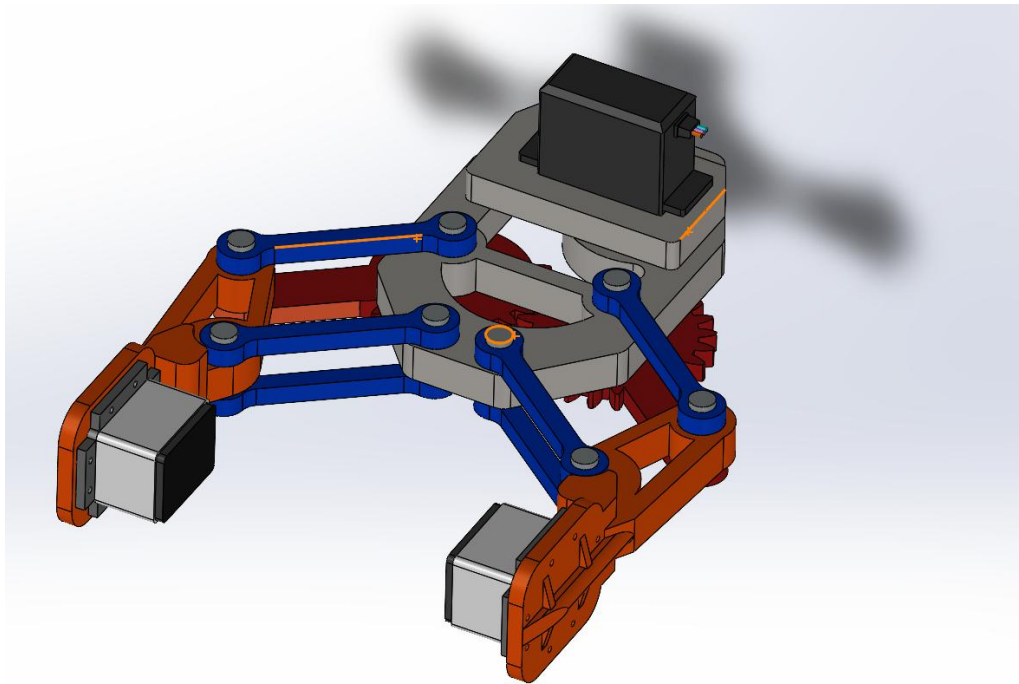


Figure 7.1 Simplified SolidWorks assembly model of the self-developed two-finger gripper and vision-based tactile sensors.

7.4 Limitations and discussion

The design involves three main trade-offs. Moving the decoder outside the sensor head can save internal space, but you need to set up an independent protected flexible cable wiring path. Although the soft G5 layer can enhance the visibility of local deformation, it will increase the sensitivity to changes in thickness, bubbles and viscoelasticity. Manual 63-point calibration can effectively deal with serious distortion and dark angle problems, but it will introduce point positioning errors, and if the camera moves, the calibration process must be re-executed.

The parameters that have not been measured include size and layer thickness changes, optical uniformity, bubble content, reassembly repeatability, and intra-surface error and depth error relative to independent reference geometry. Screw reconstruction can only confirm the visibility of features. Transfer learning still lacks real synchronised image–FT300 data and an independent test set; the gripper remains a SolidWorks assembly without a physical connection, grasping or tactile-control test.

Chapter 8 Conclusions

8.1 Main conclusions

Two compact vision-based tactile sensor prototypes were completed. The compact camera, separate decoder board, eight-LED PCB, printed enclosure, camera-locating structure and protected cable routing were integrated, and a detachable mould was used to fabricate the G50/G5/G25 interface. Manual 63-point selection, RBF rectification, sphere-based depth calibration and pixel-to-depth mapping adapted the assembled optical geometry. Screw reconstruction retained the overall profile and repeated threads, providing qualitative validation. The sensor–FT300 platform established a load path, and a 9DTact-initialised target-domain adaptation study was defined; six-axis prediction accuracy was not evaluated because a complete synchronised dataset is unavailable.

8.2 Engineering contributions and project summary

The engineering contribution is the practical adaptation of the sensing principle to the available components. The camera-locating structure and protected cable routing resolve packaging constraints; detachable tooling turns the optical and mechanical roles of three silicones into an executable staged-fabrication process; and the manual 63-point interface permits RBF rectification under strong distortion and non-uniform illumination. The two prototypes, fabrication records, calibration program, screw reconstruction and FT300 platform are the completed outcomes. The project completed the planned hardware, fabrication, calibration and platform-integration tasks.

References

- [1] R. A. Brooks, “Intelligence without Representation,” *Artificial Intelligence*, vol. 47, nos. 1–3, pp. 139–159, 1991, doi: 10.1016/0004-3702(91)90053-M.
- [2] R. Pfeifer and J. Bongard, *How the Body Shapes the Way We Think: A New View of Intelligence*. Cambridge, MA, USA: MIT Press, 2006.
- [3] A. Billard and D. Kragic, “Trends and Challenges in Robot Manipulation,” *Science*, vol. 364, no. 6446, 2019, doi: 10.1126/science.aat8414.
- [4] A. Bicchi, “Hands for Dexterous Manipulation and Robust Grasping: A Difficult Road toward Simplicity,” *IEEE Transactions on Robotics and Automation*, vol. 16, no. 6, pp. 652–662, 2000, doi: 10.1109/70.897777.
- [5] R. S. Dahiya, G. Metta, M. Valle, and G. Sandini, “Tactile Sensing—From Humans to Humanoids,” *IEEE Transactions on Robotics*, vol. 26, no. 1, pp. 1–20, 2010, doi: 10.1109/TRO.2009.2033627.
- [6] S. Luo, J. Bimbo, R. Dahiya, and H. Liu, “Robotic Tactile Perception of Object Properties: A Review,” *Mechatronics*, vol. 48, pp. 54–67, 2017, doi: 10.1016/j.mechatronics.2017.11.002.
- [7] Z. Kappassov, J.-A. Corrales, and V. Perdereau, “Tactile Sensing in Dexterous Robot Hands—Review,” *Robotics and Autonomous Systems*, vol. 74, pp. 195–220, 2015, doi: 10.1016/j.robot.2015.07.015.
- [8] A. Alfadhel and J. Kosel, “Magnetic Nanocomposite Cilia Tactile Sensor,” *Advanced Materials*, vol. 27, no. 47, pp. 7888–7892, 2015, doi: 10.1002/adma.201504015.
- [9] M. K. Johnson and E. H. Adelson, “Retrographic Sensing for the Measurement of Surface Texture and Shape,” in *2009 IEEE Conference on Computer Vision and Pattern Recognition*, pp. 1070–1077, 2009, doi: 10.1109/CVPR.2009.5206534.
- [10] W. Yuan, S. Dong, and E. H. Adelson, “GelSight: High-Resolution Robot Tactile Sensors for Estimating Geometry and Force,” *Sensors*, vol. 17, no. 12, Art. no. 2762, 2017, doi: 10.3390/s17122762.
- [11] S. Dong, W. Yuan, and E. H. Adelson, “Improved GelSight Tactile Sensor for Measuring Geometry and Slip,” in *2017 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 137–144, 2017, doi: 10.1109/IROS.2017.8202149.
- [12] E. Donlon, S. Dong, M. Liu, J. Li, E. H. Adelson, and A. Rodriguez, “GelSlim: A High-Resolution, Compact, Robust, and Calibrated Tactile-Sensing Finger,” in *2018 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 1927–1934, 2018, doi: 10.1109/IROS.2018.8593661.
- [13] I. H. Taylor, S. Dong, and A. Rodriguez, “GelSlim 3.0: High-Resolution Measurement of Shape, Force and Slip in a Compact Tactile-Sensing Finger,” in *2022 International Conference on Robotics and Automation (ICRA)*, pp. 10781–10787, 2022, doi: 10.1109/ICRA46639.2022.9811832.
- [14] M. Lambeta et al., “DIGIT: A Novel Design for a Low-Cost Compact High-Resolution Tactile Sensor with Application to In-Hand Manipulation,” *IEEE Robotics and Automation Letters*, vol. 5, no. 3, pp. 3838–3845, 2020, doi: 10.1109/LRA.2020.2977257.
- [15] B. Ward-Cherrier et al., “The TacTip Family: Soft Optical Tactile Sensors with 3D-Printed Biomimetic Morphologies,” *Soft Robotics*, vol. 5, no. 2, pp. 216–227, 2018, doi: 10.1089/soro.2017.0052.
- [16] A. Yamaguchi and C. G. Atkeson, “Combining Finger Vision and Optical Tactile Sensing: Reducing and Handling Errors while Cutting Vegetables,” in *2016 IEEE-RAS 16th International Conference on Humanoid Robots (Humanoids)*, pp. 1045–1051, 2016, doi: 10.1109/HUMANOIDS.2016.7803400.
- [17] R. Bhirangi, T. Hellebrekers, C. Majidi, and A. Gupta, “ReSkin: Versatile, Replaceable, Lasting Tactile Skins,” in *Proceedings of the 5th Conference on Robot Learning*, vol. 164, pp. 587–597, 2022.
- [18] C. Lin, Z. Lin, S. Wang, and H. Xu, “DTact: A Vision-Based Tactile Sensor that Measures High-Resolution 3D Geometry Directly from Darkness,” in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 10359–10366, 2023, doi: 10.1109/ICRA48891.2023.10160796.
- [19] C. Lin, H. Zhang, J. Xu, L. Wu, and H. Xu, “9DTact: A Compact Vision-Based Tactile Sensor for Accurate 3D Shape Reconstruction and Generalizable 6D Force Estimation,” *IEEE Robotics and Automation Letters*, vol. 9, no. 2, pp. 923–930, 2024, doi: 10.1109/LRA.2023.3339397.
- [20] G. Zhang, Y. Du, H. Yu, and M. Y. Wang, “DelTact: A Vision-Based Tactile Sensor Using a Dense Color Pattern,” *IEEE Robotics and Automation Letters*, vol. 7, no. 4, pp. 10778–10785, 2022, doi: 10.1109/LRA.2022.3196141.
- [21] W. K. Do, B. Jurewicz, and M. Kennedy, “DenseTact 2.0: Optical Tactile Sensor for Shape and Force Reconstruction,” in *2023 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 12549–12555, 2023, doi: 10.1109/ICRA48891.2023.10161150.
- [22] R. J. Woodham, “Photometric Method for Determining Surface Orientation from Multiple Images,” *Optical Engineering*, vol. 19, no. 1, 1980, doi: 10.1117/12.7972479.
- [23] D. Ma, E. Donlon, S. Dong, and A. Rodriguez, “Dense Tactile Force Estimation Using GelSlim and Inverse FEM,” in *2019 International Conference on Robotics and Automation (ICRA)*, pp. 5418–5424, 2019, doi: 10.1109/ICRA.2019.8794113.

- [24] Y. Zhang, Z. Kan, Y. Yang, Y. A. Tse, and M. Y. Wang, “Effective Estimation of Contact Force and Torque for Vision-Based Tactile Sensors with Helmholtz–Hodge Decomposition,” *IEEE Robotics and Automation Letters*, vol. 4, no. 4, pp. 4094–4101, 2019, doi: 10.1109/LRA.2019.2930477.
- [25] R. Calandra et al., “More Than a Feeling: Learning to Grasp and Regrasp Using Vision and Touch,” *IEEE Robotics and Automation Letters*, vol. 3, no. 4, pp. 3300–3307, 2018, doi: 10.1109/LRA.2018.2852779.
- [26] S. Dong, D. Ma, E. Donlon, and A. Rodriguez, “Maintaining Grasps within Slipping Bounds by Monitoring Incipient Slip,” in *2019 International Conference on Robotics and Automation (ICRA)*, pp. 3818–3824, 2019, doi: 10.1109/ICRA.2019.8793538.
- [27] J. W. James and N. F. Lepora, “Slip Detection for Grasp Stabilization with a Multifingered Tactile Robot Hand,” *IEEE Transactions on Robotics*, vol. 37, no. 2, pp. 506–519, 2021, doi: 10.1109/TRO.2020.3031245.
- [28] G. Huang, Z. Liu, L. van der Maaten, and K. Q. Weinberger, “Densely Connected Convolutional Networks,” in *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4700–4708, 2017, doi: 10.1109/CVPR.2017.243.
- [29] Robotiq Inc., “FT 300 Force Torque Sensor: Instruction Manual,” 2019. [Online]. Available: https://assets.robotiq.com/website-assets/support_documents/document/FT_Sensor_Manual_general_PDF_20190625.pdf. [Accessed: 28 Jul. 2026].
- [30] C. Lin, H. Zhang, J. Xu, L. Wu, and H. Xu, “9DTact: Official Open-Source Hardware, Software, Dataset and Pretrained Models.” [Online]. Available: <https://github.com/linchangyi1/9DTact>. [Accessed: 28 Jul. 2026].
- [31] C. Sferrazza and R. D’Andrea, “Transfer learning for vision-based tactile sensing,” in *2019 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 7961–7967, 2019, doi: 10.1109/IROS40897.2019.8967571.
- [32] Z. Chen, N. Ou, J. Jiang, and S. Luo, “Deep Domain Adaptation Regression for Force Calibration of Optical Tactile Sensors,” in *2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pp. 13561–13568, 2024, doi: 10.1109/IROS58592.2024.10801904.
- [33] Z. Chen, N. Ou, X. Zhang, and S. Luo, “TransForce: Transferable Force Prediction for Vision-Based Tactile Sensors with Sequential Image Translation,” in *2025 IEEE International Conference on Robotics and Automation (ICRA)*, pp. 237–243, 2025, doi: 10.1109/ICRA55743.2025.11127381.